

Name: Vakkalanka Sai Bhaskara Sujit
Roll No:2520090085

Skill week – 4

Step 1 – Create the Customer Table

The screenshot shows the MySQL Workbench interface. The 'Query 1' window contains the following SQL code:

```
1 CREATE TABLE Customer (  
2     Customer_ID INT PRIMARY KEY,  
3     Customer_Name VARCHAR(100),  
4     Phone VARCHAR(15),  
5     Email VARCHAR(100)  
6 );  
7  
8 INSERT INTO Customer  
9 (Customer_ID, Customer_Name, Phone, Email)  
10 VALUES  
11 (101, 'Rahul', '9876543210', 'rahul@gmail.com'),  
12 (102, 'Priya', '9876543211', 'priya@gmail.com'),  
13 (103, 'Arun', '9876543212', 'arun@gmail.com'),  
14 (104, 'Sneha', '9876543213', 'sneha@gmail.com'),  
15 (105, 'Kiran', '9876543214', 'kiran@gmail.com');
```

The 'Output' window shows the execution results:

#	Time	Action	Message
1	19:54:36	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100), Phone ...	0 row(s) affected
2	19:54:36	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email) VALUES (101, 'Rahul', '9876543210', 'ra...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

Step 2 – Verify the Customer Table

The screenshot shows the MySQL Workbench interface. The 'Query 1' window contains the following SQL code:

```
3 Customer_ID INT,  
4 Account_Type VARCHAR(20),  
5 Balance DECIMAL(12,2),  
6 Branch VARCHAR(50),  
7 FOREIGN KEY (Customer_ID)  
8 REFERENCES Customer(Customer_ID)  
9 );  
10  
11 SELECT * FROM Account;
```

The 'Output' window shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	19:54:36	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100), Phone...	0 row(s) affected	0.031 sec
2	19:54:36	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email) VALUES (101, 'Rahul', '9876543210', 'ra...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
3	20:08:22	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR...	0 row(s) affected	0.031 sec
4	20:08:22	SELECT * FROM Account LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec

Step 3 – Create the Loan Table

The screenshot shows the MySQL Workbench interface. In the 'Query' tab, the following SQL code is entered:

```
2 Loan_ID INT PRIMARY KEY,  
3 Customer_ID INT,  
4 Loan_Type VARCHAR(30),  
5 Loan_Amount DECIMAL(12,2),  
6 Interest_Rate DECIMAL(5,2),  
7 FOREIGN KEY (Customer_ID)  
8 REFERENCES Customer(Customer_ID)  
9 );  
10 SHOW TABLES;
```

The 'Result Grid' shows the output of the 'SHOW TABLES' command, listing the tables: 'Tables_in_skillweek4', 'account', 'customer', and 'loan'.

The 'Output' tab shows the execution results of the SQL statements:

#	Time	Action	Message	Duration / Fetch
1	19:54:36	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100), Phone...	0 row(s) affected	0.031 sec
2	19:54:36	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email) VALUES (101, 'Rahul', '9076543210', 'r...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
3	20:08:22	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR...	0 row(s) affected	0.031 sec
4	20:08:22	SELECT * FROM Account LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
5	20:09:46	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30),	0 row(s) affected	0.047 sec
6	20:10:21	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30),	Error Code: 1050, Table 'loan' already exists	0.000 sec
7	20:10:31	SHOW TABLES	3 row(s) returned	0.016 sec / 0.000 sec

Step 4 – Create Bank_Transaction Table and Insert Account & Transaction Records

The screenshot shows the MySQL Workbench interface. In the 'Query' tab, the following SQL code is entered:

```
5 Amount DECIMAL(12,2),  
6 FOREIGN KEY (Account_No)  
7 REFERENCES Account(Account_No)  
8 );  
9 INSERT INTO Account  
10 (Account_No, Customer_ID, Account_Type, Balance, Branch)  
11 VALUES  
12 (10001, 101, 'Savings', 50000, 'Hyderabad'),  
13 (10002, 102, 'Savings', 75000, 'Vijayawada'),  
14 (10003, 103, 'Current', 120000, 'Bangalore'),  
15 (10004, 104, 'Savings', 45000, 'Chennai');
```

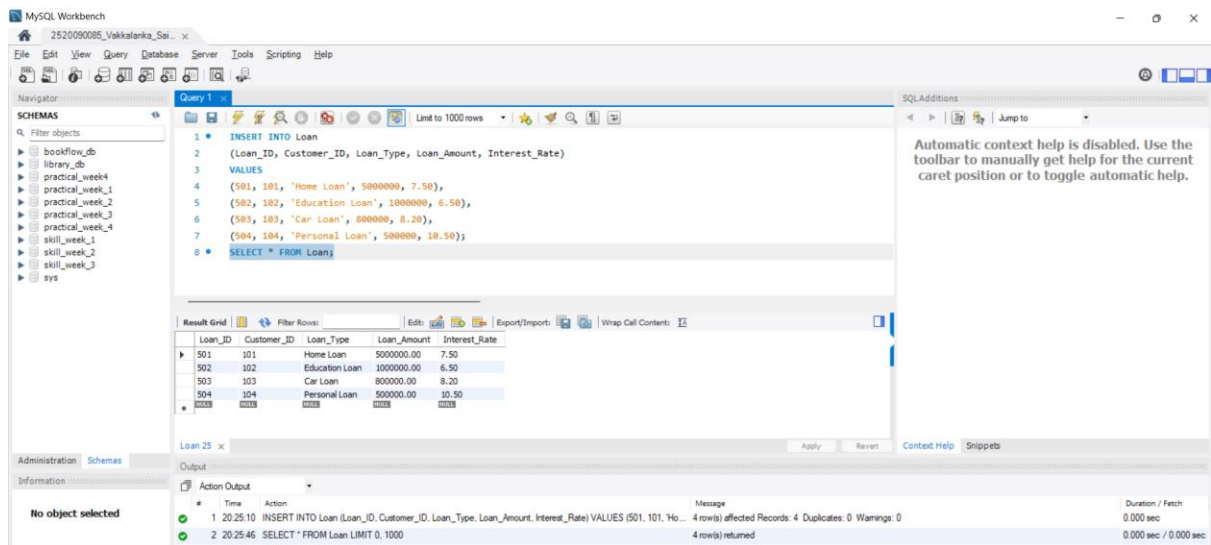
The 'Result Grid' shows the output of the 'INSERT INTO Account' command, listing the accounts:

Transaction_ID	Account_No	Transaction_Type	Amount
1	10001	DEPOSIT	10000.00
2	10002	DEPOSIT	15000.00
3	10003	WITHDRAW	20000.00
4	10004	DEPOSIT	5000.00
5	10005	WITHDRAW	10000.00

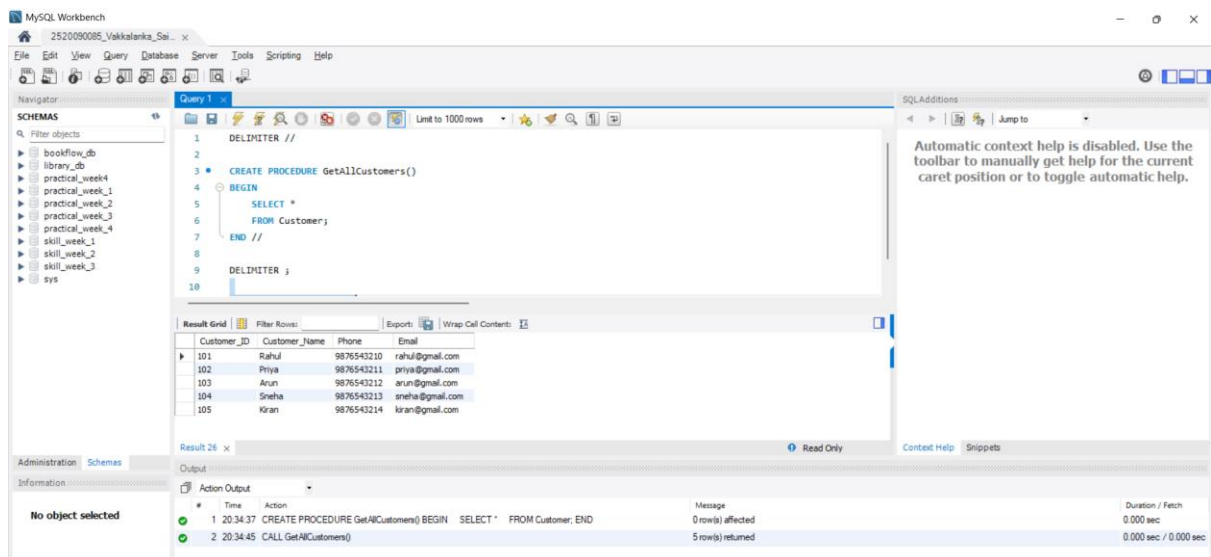
The 'Output' tab shows the execution results of the SQL statements:

#	Time	Action	Message	Duration / Fetch
1	20:11:47	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account...	0 row(s) affected	0.032 sec
2	20:12:29	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'S...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.063 sec
3	20:12:46	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, DEPOSIT, 1000...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
4	20:12:46	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
5	20:12:59	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

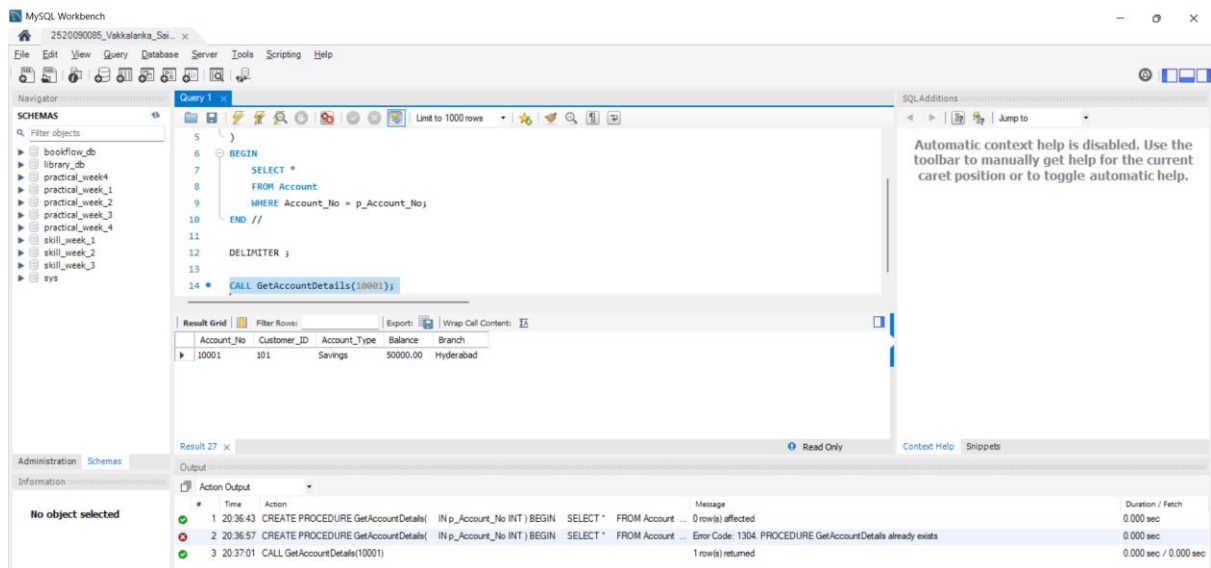
Step 5 – Insert Loan Records



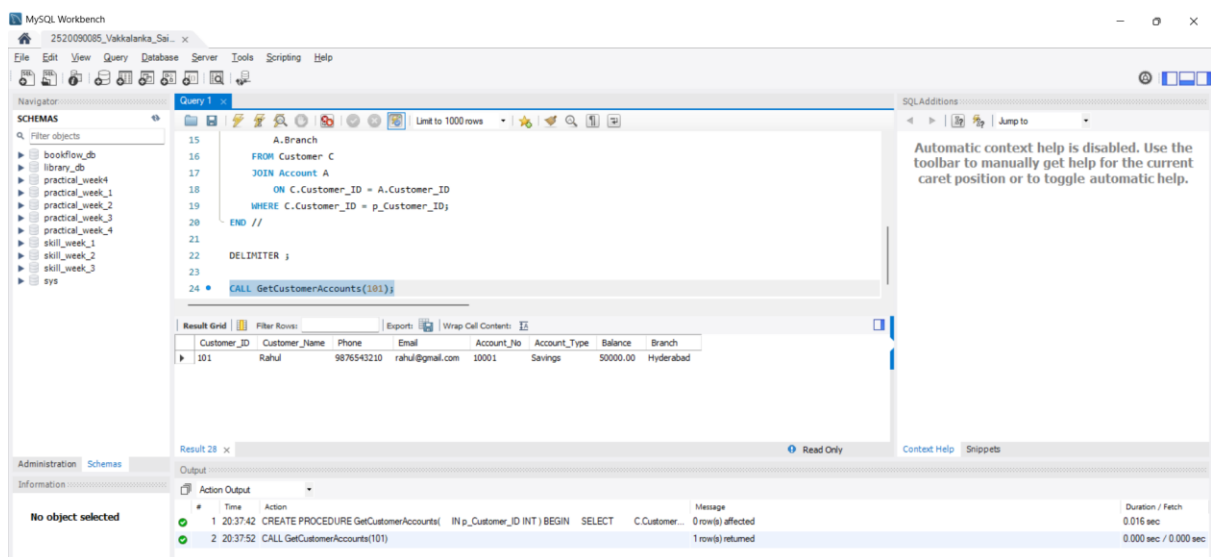
Step 6 – Create GetAllCustomers()



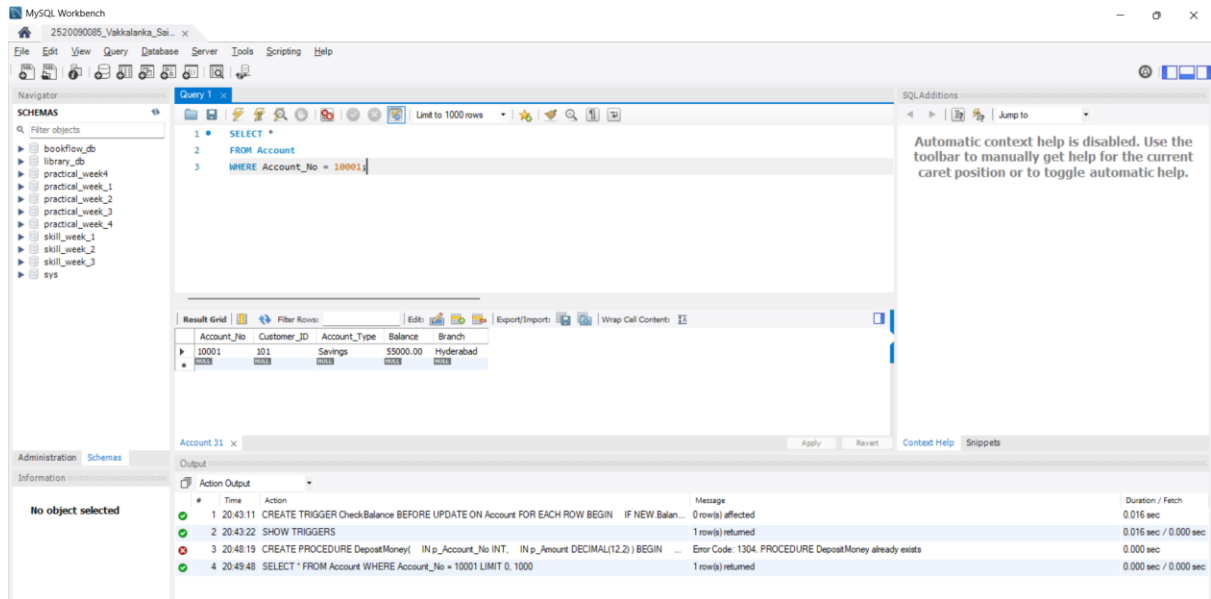
Step 7 – Create and Execute GetAccountDetails()



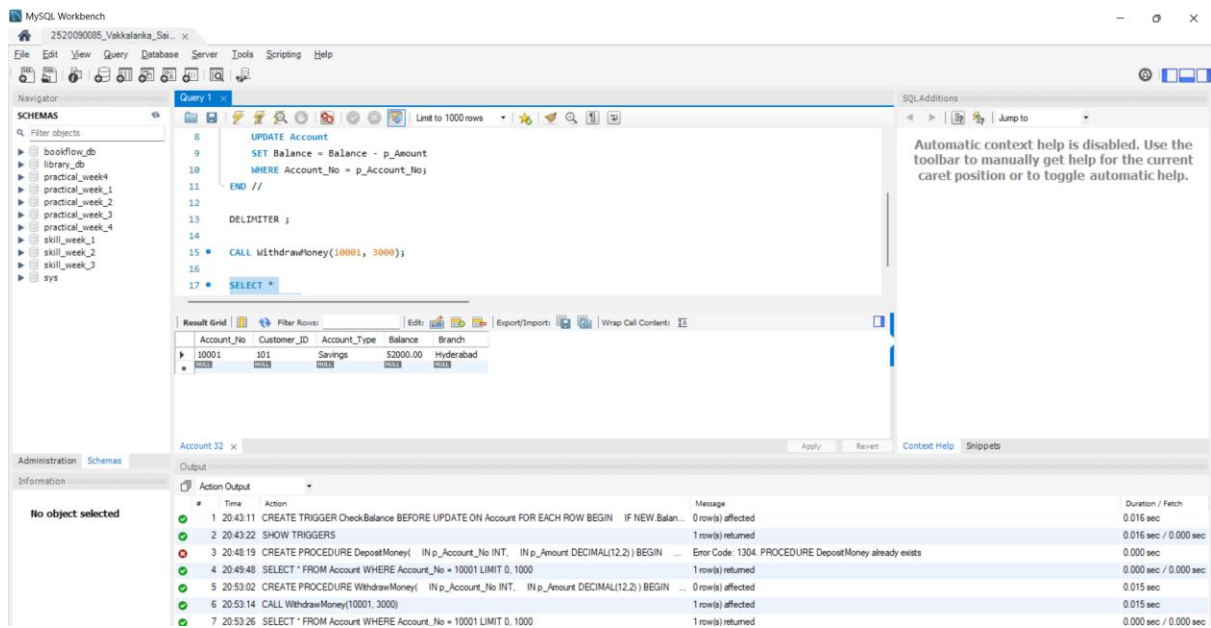
Step 8 – Create GetCustomerAccounts()



Step 9 – Create and execute GetAllCustomers() to retrieve all customer details



Step 10 – Create and execute GetAccountDetails()



Step 11 – Create GetCustomerLoans()

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

- bookflow_db
- library_db
- practical_week4
- practical_week_1
- practical_week_2
- practical_week_3
- practical_week_4
- skill_week_1
- skill_week_2
- skill_week_3
- sys

Query 1

```

12      L.Interest_Rate
13      FROM Customer C
14      JOIN Loan L
15      ON C.Customer_ID = L.Customer_ID
16      WHERE C.Customer_ID = p_Customer_ID;
17  END //
18
19  DELIMITER ;
20
21

```

Result Grid

Customer_Name	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
Rahul	501	Home Loan	500000.00	7.50

Result 34

Read Only

Context Help Snippets

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Administration Schemas

Information

No object selected

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	20:53:02	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN ...	0 row(s) affected	0.015 sec
6	20:53:14	CALL WithdrawMoney(10001, 3000)	1 row(s) affected	0.015 sec
7	20:53:26	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
8	20:54:16	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW Bal...	Error Code: 1359. Trigger already exists	0.000 sec
9	20:54:25	SHOW TRIGGERS	1 row(s) returned	0.000 sec / 0.000 sec
10	20:55:12	CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT) BEGIN SELECT C.Customer_...	0 row(s) affected	0.016 sec
11	20:55:29	CALL GetCustomerLoans(101)	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Step 12 – Create HighBalanceAccounts()

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

- bookflow_db
- library_db
- practical_week4
- practical_week_1
- practical_week_2
- practical_week_3
- practical_week_4
- skill_week_1
- skill_week_2
- skill_week_3
- sys

Query 1

```

6      BEGIN
7      SELECT *
8      FROM Account
9      WHERE Balance >= MinimumBalance
10     ORDER BY Balance DESC;
11  END //
12
13  DELIMITER ;
14
15  CALL HighBalanceAccounts(50000);

```

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10003	103	Current	120000.00	Bangalore
10005	105	Current	98000.00	Hyderabad
10002	102	Savings	75000.00	Vijayanada
10001	101	Savings	52000.00	Hyderabad

Result 35

Read Only

Context Help Snippets

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Administration Schemas

Information

No object selected

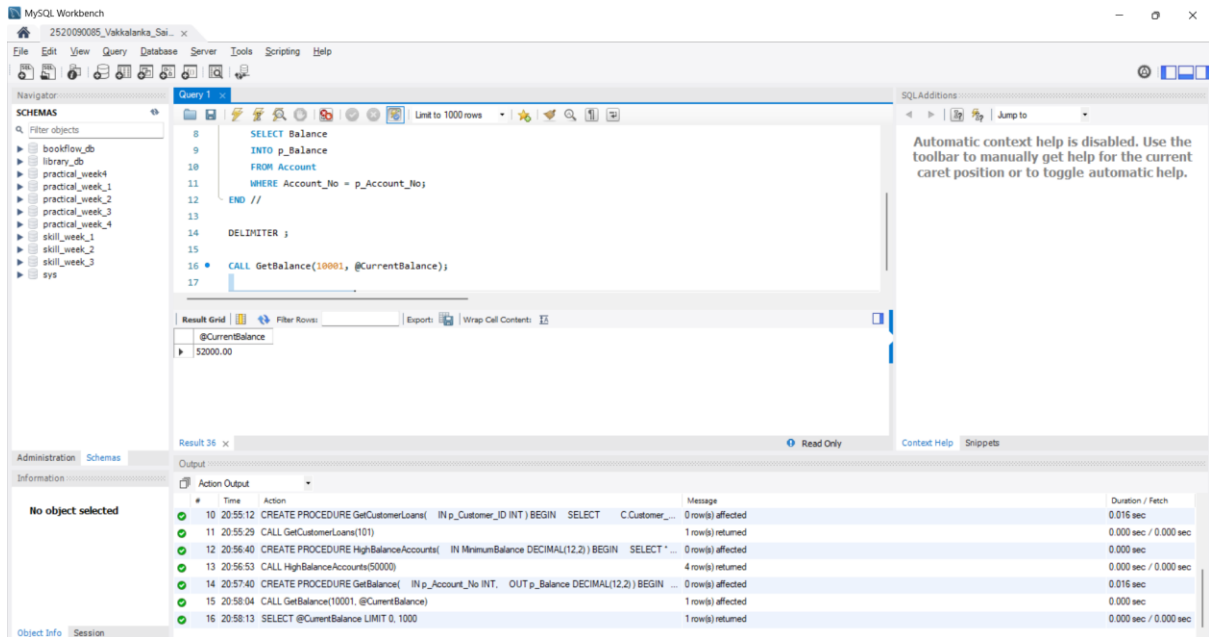
Output

Action Output

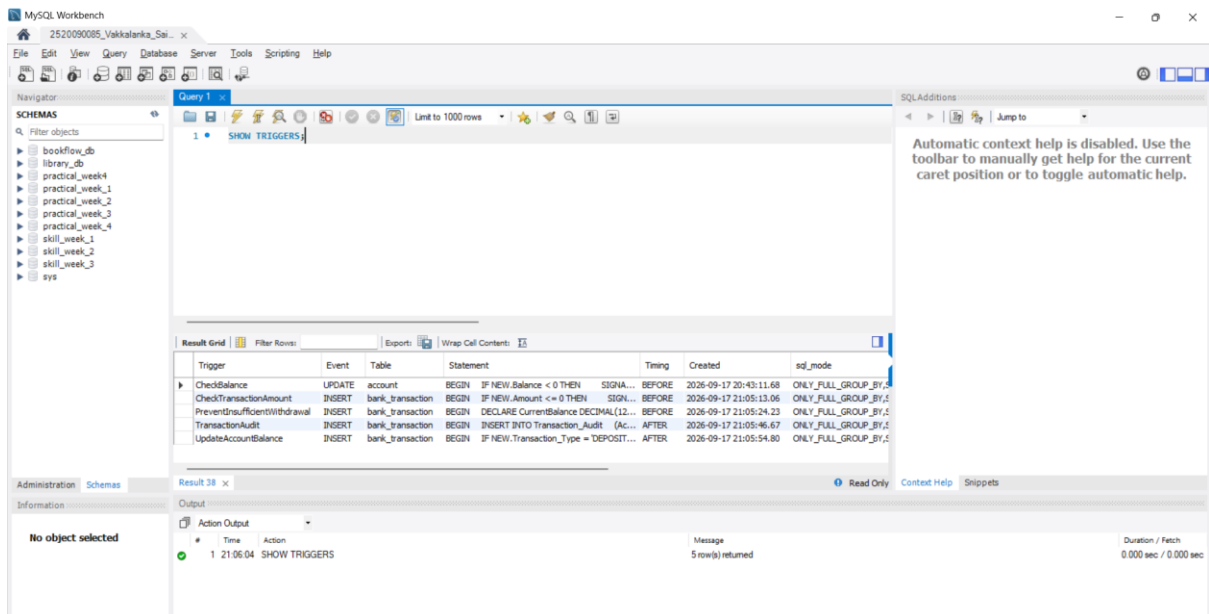
#	Time	Action	Message	Duration / Fetch
7	20:53:26	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
8	20:54:16	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW Bal...	Error Code: 1359. Trigger already exists	0.000 sec
9	20:54:25	SHOW TRIGGERS	1 row(s) returned	0.000 sec / 0.000 sec
10	20:55:12	CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT) BEGIN SELECT C.Customer_...	0 row(s) affected	0.016 sec
11	20:55:29	CALL GetCustomerLoans(101)	1 row(s) returned	0.000 sec / 0.000 sec
12	20:56:40	CREATE PROCEDURE HighBalanceAccounts(IN MinimumBalance DECIMAL(12,2)) BEGIN SELECT *	0 row(s) affected	0.000 sec
13	20:56:53	CALL HighBalanceAccounts(50000)	4 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Step 13 – Create GetBalance() to Check Account Balance



Step 14 – Create GetCustomerLoans() to retrieve loans belonging to a selected customer



Step 15 – Create GetBranchAccounts()

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- library_db
- practical_week4
- practical_week_1
- practical_week_2
- practical_week_3
- practical_week_4
- skill_week_1
- skill_week_2
- skill_week_3
- sys

Query 1

```

3 CREATE PROCEDURE GetBranchAccounts(
4     IN p_Branch VARCHAR(50)
5 )
6 BEGIN
7     SELECT *
8     FROM Account
9     WHERE Branch = p_Branch;
10 END //
11
12 DELIMITER ;
13

```

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	52000.00	Hyderabad
10005	105	Current	98000.00	Hyderabad

Administration Schemas

Information

No object selected

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	21:06:04	SHOW TRIGGERS	5 row(s) returned	0.000 sec / 0.000 sec
2	21:07:11	CREATE PROCEDURE GetBranchAccounts(IN p_Branch VARCHAR(50)) BEGIN SELECT * FROM Ac...	0 row(s) affected	0.015 sec
3	21:07:21	CREATE PROCEDURE GetBranchAccounts(IN p_Branch VARCHAR(50)) BEGIN SELECT * FROM Ac...	Error Code: 1304. PROCEDURE GetBranchAccounts already exists	0.000 sec
4	21:07:25	CALL GetBranchAccounts(Hyderabad)	2 row(s) returned	0.000 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Step 16 – Create GetCustomerFullDetails()

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- library_db
- practical_week4
- practical_week_1
- practical_week_2
- practical_week_3
- practical_week_4
- skill_week_1
- skill_week_2
- skill_week_3
- sys

Query 1

```

13     A.Balance,
14     A.Branch
15 FROM Customer C
16 JOIN Account A
17 ON C.Customer_ID = A.Customer_ID
18 WHERE C.Customer_ID = p_Customer_ID;
19 END //
20
21 DELIMITER ;
22
23 CALL GetCustomerFullDetails(101);

```

Result Grid

Customer_Name	Phone	Email	Account_No	Account_Type	Balance	Branch
Rahul	9878543210	rahul@gmail.com	10001	Savings	52000.00	Hyderabad

Administration Schemas

Information

No object selected

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	21:06:04	SHOW TRIGGERS	5 row(s) returned	0.000 sec / 0.000 sec
2	21:07:11	CREATE PROCEDURE GetBranchAccounts(IN p_Branch VARCHAR(50)) BEGIN SELECT * FROM Ac...	0 row(s) affected	0.015 sec
3	21:07:21	CREATE PROCEDURE GetBranchAccounts(IN p_Branch VARCHAR(50)) BEGIN SELECT * FROM Ac...	Error Code: 1304. PROCEDURE GetBranchAccounts already exists	0.000 sec
4	21:07:25	CALL GetBranchAccounts(Hyderabad)	2 row(s) returned	0.000 sec / 0.000 sec
5	21:08:03	CREATE PROCEDURE GetCustomerFullDetails(IN p_Customer_ID INT) BEGIN SELECT C.Customer...	0 row(s) affected	0.016 sec
6	21:08:14	CALL GetCustomerFullDetails(101)	1 row(s) returned	0.000 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Step 17 – Create GetCustomerFullDetails() and GetTotalCustomerBalance()

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- library_db
- practical_week4
- practical_week_1
- practical_week_2
- practical_week_3
- practical_week_4
- skill_week_1
- skill_week_2
- skill_week_3
- sys

Query 1: x

1 • CALL GetTotalCustomerBalance(101);

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Customer_ID	Total_Balance
101	52000.00

Filter Rows: Exports: Wrap Cell Contents: X

Administration Schemas

Result 41 x

Read Only Context Help Snippets

Information

No object selected

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	21:09:47	CALL GetTotalCustomerBalance(101)	1 row(s) returned	0.000 sec / 0.000 sec